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After completing the preparation materials, students should be able to:

1. **Biochemistry / Physiology foundation of NSAID pharmacology:** Explain the synthesis of the biologically active prostanoids, PGE2, PGI2, PGF2α, PGD2, and TXA2, and their diverse receptor-mediated actions in various tissues.
2. **NSAIDs Class Pharmacokinetics:** Identify the NSAIDs class pharmacokinetic properties that are relevant for optimizing patients care.
3. **Mechanism of action:** Describe how inhibiting cyclooxygenase reduces the synthesis of biologically active prostanoids.
4. **Therapeutic uses:** Correlate the NSAID therapeutic uses with their relative affinities for COX-1- and COX-2 using specific examples.
5. **Adverse effects:** Link the NSAIDs' potentially serious adverse effects to the inhibition of COX-1 and COX-2 mediated prostanoid activity in the various tissues using specific examples.
6. **Drug interactions and Precautions:** Predict the potential drug interactions and precautions related to NSAID therapy, including in special populations (pregnancy and lactation, pediatric and geriatric patients, and patients with comorbidities).

NSAIDs Pharmacology Study Hints

<u>First</u>
The Big Picture: Class Effects

<u>Then</u>
How is each drug the same and different? Drug-specific Effects

1. Become familiar with the **physiological foundation** of the pharmacology.
2. Then, verbalize the **common thread** that stitches together all drugs in the group.
3. Then, learn the NSAIDs **CLASS** mechanisms / effects.
   - Pharmacokinetics
   - Mechanisms of action, mechanisms of adverse effects, mechanisms of drug interactions
   - Therapeutic uses
   - Drug interactions: drug-disease, drug-procedure, drug-drug
   - Contraindications – patient-specific effects related to comorbidities and drug interactions
4. **Then**, identify the starred (✳) drugs of each subclass and how they represent the class effects. (Make the drug names your friends by saying them out loud.)
5. **Finally**, identify the drug-specific features of each drug:

aspirin, salicylates, non-aspirin NSAIDs, acetaminophen

Key points

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The **biochemistry and physiology** of arachidonic acid (AA) and conversion to the eicosanoids, the actions of the cyclooxygenases (COXs), the synthesis of prostanoids and their effects are **ESSENTIAL** for understanding the pharmacology of the NSAIDs.

Prostanoids: Subclass of potent autocrine / paracrine signaling molecules derived from AA

Prostaglandins: PGE2, PGF2α, PGD2

Other Prostanoids: PGI2 (prostacyclin) and TXA2 (thromboxane A2)

- Produced in minute quantities by virtually all tissues
- Act locally on cells that secrete them or cells near the site of synthesis
- Brief action (~90 seconds)
- Bind various types of G protein-coupled receptors $\rightarrow$ specific functions, which depend on the tissue, enzyme pathways within the tissue, and receptors that are activated.

NSAIDs are chemically dissimilar compounds that inhibit COX-1 and/or COX-2.

- Decrease prostanoid synthesis
- Therapeutic uses for analgesic, anti-inflammatory, and antipyretic effects
- COX inhibition also results in potentially serious adverse effects due to prevention of constitutive and inducible effects of prostanoids.

COX-1-derived Prostanoids → Homeostatic Functions of Prostanoids predominantly constitutive actions

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TAKEAWAY: NSAID use can disrupt the protective homeostatic functions of prostanoids leading to potentially serious adverse effects in at-risk individuals.

Several NSAIDs are available without a prescription – people self-treat.

GI tract cytoprotection PGE2 / PGI2	Stimulate mucus and bicarbonate secretion Maintain mucosal blood flow Reduce acid secretion Promote epithelial repair and turnover	Immune system PGD2	Maintains local tissue immune homeostasis Regulates mast cell function
Platelet function TXA2	Promote platelet aggregation Enhances platelet activation Causes local vasoconstriction Helps maintain homeostatic platelet plug	Vascular tone PGE2 / PGI2	Basal hemodynamics to support blood pressure: Contribute to basal vasodilation Counteract excessive vasoconstriction Support endothelial hemostasis
Renal autoregulation PGE2 / PGI2	Vasodilation of afferent arteriole Maintenance of glomerular filtration rate (GFR) Promote sodium and water excretion Renin release baseline regulation	Reproductive homeostasis	PGF2α uterine smooth muscle contraction (menstrual cycle endometrial changes) Initiation and maintenance of labor and supports postpartum involution
CNS PGE2 / PGD2	Microglial homeostasis Sleep regulation		

Check your knowledge

What determines whether certain prostanoids are constitutively expressed or inducible? (and define the terms)

How do the prostanoids exert their different effects in various body tissues?

Pharmacokinetics: Class Properties

Chemically heterogeneous; **weak acids** (pka typically 3-5)
(except nabumetone, a nonacidic prodrug)

A	Rapid, complete, not affected by food onset of analgesia 30-60 minutes; T_{max} 1-4 hrs
D	~99% albumin bound; cross blood-brain barrier, accumulate at sites of inflammation and synovial fluid

rapid pain relief

accumulate at site of action

Analgesic effects are related to CNS actions: reduced central sensitization, modulation of descending pain pathways, and possibly interaction with other systems. Accumulation in synovial fluid is directly related to NSAIDs' effectiveness in arthritis.

M	Hepatic, minor substrates of CYPs, esp CYP3A, CYP2C families; glucuronidation; enterohepatic circulation
E	glomerular filtration / tubular secretion of metabolites (small amount unchanged drug)
t½	< 6 hours to > 10 hours

Caution impaired organ fx

short, moderate, or long durations of action

NSAIDs: Mechanism → Effect Relationship

Therapeutic effects:

- **COX-2 inhibition** → analgesic, antipyretic, and anti-inflammatory actions

Adverse effects:

- **COX-1 inhibition** → prevents synthesis of GI cytoprotective PGE2 and PGI2
- **COX-1 inhibition in platelets** → TXA2 synthesis inhibition → reduced platelet aggregation (desirable effect only with low dose aspirin)
- **COX-2 inhibition in endothelial cells** → PGI2 synthesis inhibition → inhibits counterbalancing platelet disaggregation → increased risk of bleeding AND increased risk of thrombosis
- **COX-1 and COX-2 inhibition** → renal PGE2 and PGI2 synthesis inhibition → decreases renal blood flow and sodium / water excretion (and other effects)

What is the "common thread" that ties all these drugs together?

Please lift your eyes away from the PPT slide and try to answer this question.

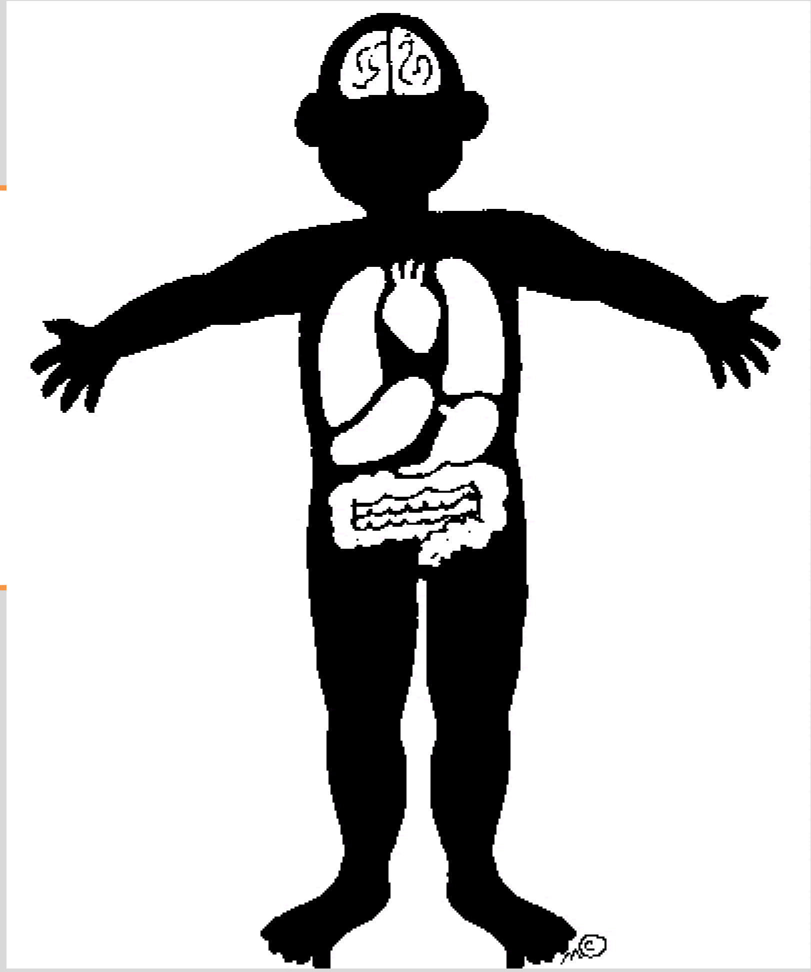
Prostanoids

signaling molecules in MANY biological processes

Therapeutic Effects

NSAIDS inhibit prostanoid synthesis

TOXICITIES



All NSAIDs treat pain, fever, and inflammation.

All NSAIDs can cause potentially serious adverse effects in vulnerable patients.

Check your knowledge

How do NSAIDs inhibit the synthesis of biologically active prostanoids?

How does the NSAIDs' mechanism of action relate to their therapeutic and adverse effects?

Why is it helpful to recognize the affinities of the individual NSAIDs for COX-1 and COX-2?

Mechanisms of cardiovascular effects

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Aspirin **(protective)**	**non-aspirin nsNSAIDs** (increased risk of thrombosis)	**Selective COX-2 inhibitors** (increased risk of thrombosis)
Low-dose aspirin inhibits platelet COX-1 mainly in the portal system. Much of the dose is metabolized on the first pass through the liver.	nsNSAIDs achieve therapeutic blood levels in the systemic circulation.	COX-2 selective NSAIDs block endothelial COX-2 in the systemic circulation but do not block platelet COX-
• This action shifts the balance towards antiplatelet effects when aspirin is taken daily.	• Block both COX-1 in platelets & COX-2 in endothelial cells. • COX-2 more sensitive to block → shift balance towards platelet thrombus formation, particularly under inflammatory/shear stress conditions	• This shifts the balance towards platelet thrombus formation.

Clinical aspects of NSAIDs-related cardiovascular risk: atherothrombosis, myocardial infarction, and stroke (except with aspirin)

Chronic or repeated NSAID use increases risk of thrombosis.

BP elevation and heart failure exacerbation contribute to CV risk.

1. Higher dose / longer duration → ↑ risk.
2. COX-2 selective and non-aspirin nsNSAIDs are both associated with increased CV risk.

PRACTICE POINTS

1. Low dose / Short duration
2. Reserve COX-2 selective drugs for high GI risk
3. Consider alternate treatment for high-risk patients

Risk factors of coronary heart disease and stroke include tobacco use, alcohol use, high blood pressure (hypertension), high cholesterol, obesity, physical inactivity, and unhealthy diets.

US Boxed Warning: All NSAIDs (except aspirin) are contraindicated during perioperative cardiopulmonary bypass graft surgery →↑ thrombotic and CV risk.

Renal Effects: First – physiology

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Nephron filters the blood; regulates fluid, electrolytes, waste

Afferent arteriole controls how much blood enters the glomerulus.

Efferent arteriole controls how much pressure builds up inside the glomerulus.

Together they fine-tune RBF and GFR.

Autoregulatory effects of prostanoids

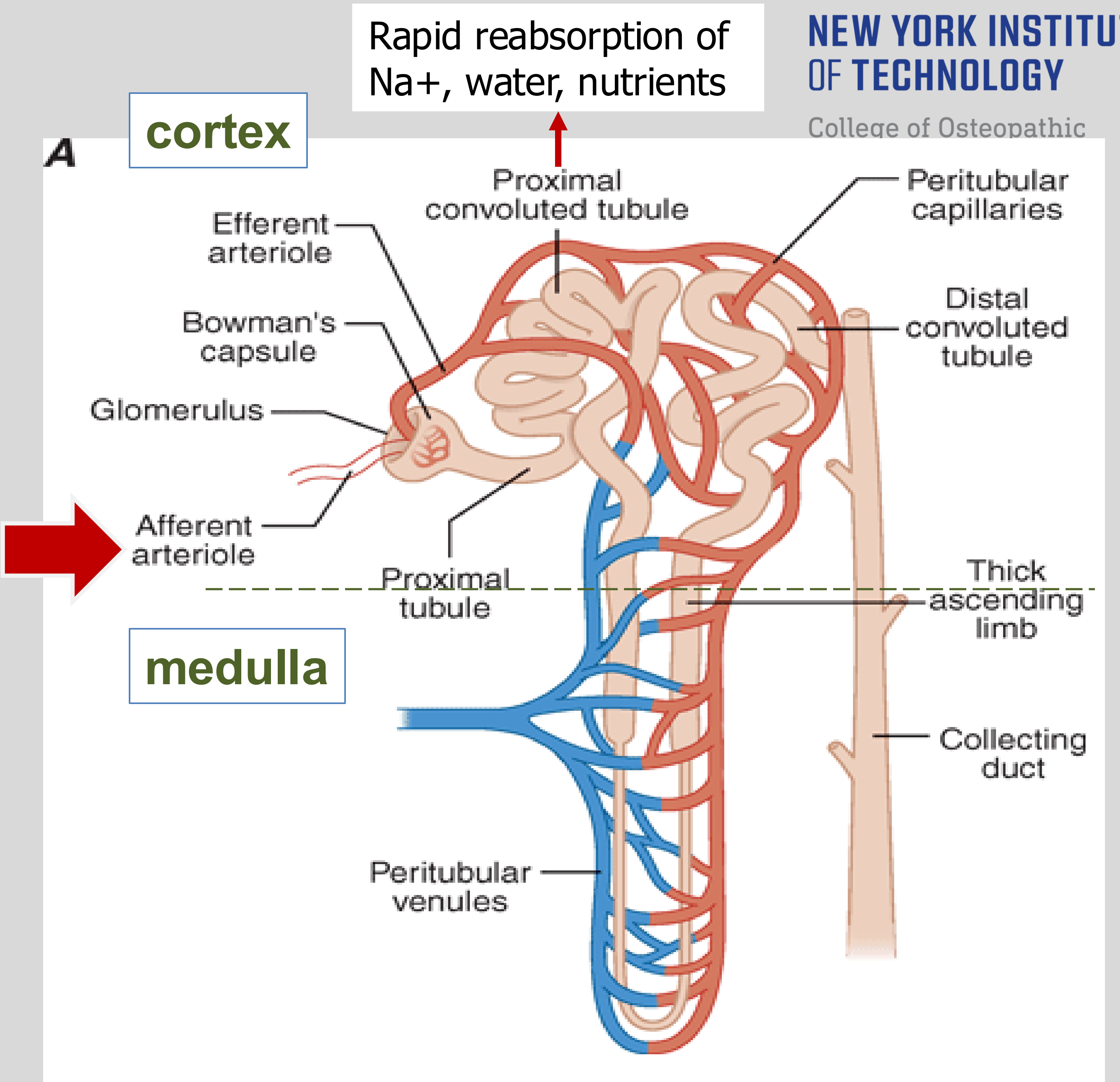
Cortical COX-2-derived PGE2 and PGI2 → arteriolar vasodilation → **maintains local RBF and GFR**

Medullary COX-2-derived PGE2 and PGI2 → ↑ **medullary blood flow and ↓ tubular Na^+ reabsorption**

COX-1-derived PGE2 → **promote salt and water excretion in collecting ducts**

PGE2-stimulated **renin release in volume-contracted states**

RBF: renal blood flow
GFR: glomerular filtration rate



Source: Longo DL, Fauci AS, Kasper DL, Hauser SL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine, 18th Edition*: www.accessmedicine.com

NSAIDs block the autoregulatory effects of PGs.

Now: Link the Prostanoids – NSAIDs Mechanism – Renal Adverse Effects

Inhibition of prostanoid synthesis →

1. Afferent arteriole vasoconstriction
   - Reduces renal perfusion
   - Worsens GFR / renal function
   - Increases proteinuria
2. Causes salt and water retention
   - Hypertensive complications (new onset or worsened control)
   - Increased risk of edema
   - Hyperkalemia (↑ plasma K^+)

Risk factors:

Prostanoid formation becomes crucial in patients with marginally functioning kidneys and volume-contracted states.

- hypovolemia
- congestive heart failure
- hepatic cirrhosis
- chronic kidney disease
- other states of activation of the sympatho-adrenal or renin-angiotensin system

The TAKEAWAY:

NSAID administration in patients with ***v***olume-contracted states and other states of activation of the sympathoadrenal or renin-angiotensin-aldosterone system (RAAS) removes the autoregulatory effects of prostanoids, resulting in a major ***reduction in RBF and GFR. Acute kidney injury*** can result.

Renin-angiotensin-aldosterone system (RAAS) increases and maintains blood pressure.

Other effects of prostaglandin inhibition

- **CNS:**

headache; dizziness; tinnitus

- **Dermatologic:**

rash; pruritis; ecchymosis; purpura

- **Hematologic:** (rare)

agranulocytosis, thrombocytopenia, aplastic anemia

Practice point

A thorough patient evaluation to identify risk factors for toxicities

- GI,
- CV,
- acute kidney injury,
- and other toxicities

can minimize NSAID adverse drug events for the individual patient.

Why focus on NSAIDs' adverse effects?

FYI: Approximately 11% of hospital admissions for adverse drug events are for nonaspirin NSAID GI, renal, neurological, and allergic adverse effects.

(Howard RL, Avery AJ, Slavenburg S, et.al. Which drugs cause preventable admissions to hospital? A systematic review. Br J Clin Pharmacol. 2007 Feb.;(63)2:136-47. Epub 2006 Jun 26)

Check your knowledge

How does NSAID use increase the risk of bleeding?

What organ system is most frequently injured by NSAID use?

Check your knowledge

Which NSAIDs subclass has a dose-dependent lower risk of GI adverse effects?

Chronic or repeated NSAID use can lead to acute kidney injury by what mechanisms?

What are the various mechanisms of cardiovascular adverse effects?

NSAIDs Class Drug-disease interactions

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🔔General Precautions🔔

- Elderly
- Patients with cardiovascular risk factors
  - Hypertension
  - Fluid retention
  - Heart failure
- Coagulation disorder
- GI ulcer / bleeding
- Severe hepatic or renal impairment

CONTRAINDICATIONS

- Asthma (see aspirin)
- Coronary artery bypass graft
  - Before, during, and immediately after
- Bariatric surgery (nsNSAIDs)
  - Reduced mucosal defenses from the surgery is magnified with NSAID use
- Pregnancy
- Hypersensitivity reaction to aspirin or any other NSAID

NSAIDs Class Drug-Drug Interactions

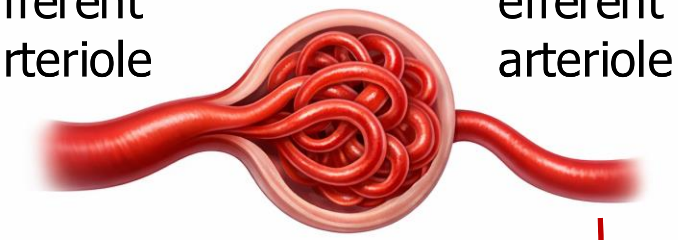
Mechanisms on the following slides

- RAS Inhibitors
  - Angiotensin converting enzyme inhibitors (ACE inhibitors)
  - Angiotensin receptor blockers (ARBs)
  - Direct renin inhibitors (DRIs)
- Diuretics
- Antihypertensive drugs

- Anticoagulants
- Antiplatelet drugs
- Drugs that cause mucosal injury
- Selective serotonin reuptake inhibitors (SSRIs)
- Lithium

Renin-Angiotensin System (RAS) inhibitors:

Angiotensin converting enzyme inhibitors (ACE inhibitors)
Angiotensin receptor blockers (ARBs)
Direct renin inhibitors (DRIs)



Copilot-generated figure

1. RAS inhibitors block ANG II's vasoconstrictor effects on the efferent arteriole
2. NSAIDs block PG's vasodilator effects on afferent arteriole

➡ decrease glomerular capillary pressure
↓
may cause acute kidney injury
In vulnerable patient

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1. RAS inhibitors cause K+ reabsorption
2. NSAIDs increase K+ reabsorption

➡ hyperkalemia → arrhythmia

NSAIDs cause Na+ and water reabsorption
↓
Increase intravascular volume

➡ Blunt antihypertensive effects of the RAS inhibitors

Mechanisms of NSAIDs Drug-Drug Interactions

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Diuretics for treatment of hypertension and edema	• Diuretics increase Na+ and water excretion. NSAIDs may attenuate this effect. • Some diuretics stimulate COX activity. NSAIDs block the synthesis of vasodilating PGs, thus reduce diuretic efficacy. • Some diuretics cause potassium reabsorption (potassium sparing diuretics) → risk of hyperkalemia with NSAIDs
Antihypertensive drugs	NSAIDs increase Na+ and water reabsorption, which can blunt the blood pressure lowering effects of the antihypertensive agents.
Anticoagulants, Antiplatelet drugs, Fibrinolytics	Increase the risk of bleeding (brain, GI tract, anywhere)

Mechanisms of NSAIDs Drug-Drug Interactions

Drugs that cause injury to the intestinal mucosa	↑ risk of gastrointestinal adverse events (alcohol; glucocorticoids; aspirin; others)
Selective serotonin reuptake inhibitors SSRIs are antidepressants	• Serotonin is taken up by platelets via serotonin transporters and stored in the platelets. • Serotonin is released from platelets (along with TXA2) and stimulates platelet aggregation. • SSRIs inhibit uptake of serotonin into platelets → ↓ amount of serotonin released from activated platelets → ↓ platelet aggregation → ↑ risk of bleeding • NSAIDs also ↑ risk of bleeding by inhibiting TXA2 synthesis.
Lithium (Li^+) (for bipolar disorder)	NSAIDs may increase Li^+ levels by increasing Li^+ reabsorption. Lithium has a narrow therapeutic window and significant toxicity.

Check your knowledge

How does maternal chronic NSAID use affect the fetus?

Sum up the mechanisms of NSAIDs-related drug interactions in two sentences.

NSAIDs Class Effects

PK: NSAIDs are chemically diverse weak acids.

- A: Well absorbed
- D: Highly bound to plasma proteins and widely distributed in tissues
- M: Mainly via CYP3A or 2C enzymes with biliary excretion and enterohepatic circulation. E: Metabolites active drug are excreted in urine
- t½: Half-lives vary from <6 hours) to >10 hours.

Therapeutic effects:

- Analgesic,
- Anti-inflammatory,
- Antipyretic

Efficacy:

- All NSAIDs have similar effectiveness in the management of mild to moderate pain, fever, and inflammation. Some patients seem to respond better to one drug than to another.

NSAIDs Class Adverse Effects

- GI: loss of cytoprotection→ dyspepsia, ulcers, bleeding, perforation
  - Risk is increased by age, glucocorticoids, anticoagulants, SSRIs, and prior ulcer
- CV:
  - Inhibition of TXA2-mediated platelet aggregation → reduced formation of platelet plug → increased risk of bleeding
  - Inhibiting PGI2-mediated platelet disaggregation allows unopposed TXA2-mediated platelet aggregation → prothrombotic effect → increased risk myocardial infarction, stroke
    - nsNSAIDs: COX2 inhibition is more sensitive than TXA2 inhibition → imbalance towards thrombus formation
    - COX-2 selective agents inhibit endothelial COX2 but not TXA2 → imbalance towards thrombus formation
- Renal: Inhibition of COX-1 and COX-2 derived prostanoids → ↓ renal perfusion and GFR, Na+ and water reabsorption (↑ blood pressure / edema →↑ risk of MI, stroke / heart failure), and hyperkalemia

Check your knowledge

What determines whether certain prostanoids are constitutively expressed or inducible? (and define the terms)

- Constitutive expression: An enzyme (or receptor) that remains in an active state (always on) without requiring a stimulus
- Inducible expression: An enzyme (or receptor) that is activated only in response to a specific stimulus
- COX-1 is constitutively active at basal levels. Prostanoids produced by COX-1 primarily provide homeostatic (housekeeping) functions. PGE2 and PGI2 are widely expressed.
- COX-2 is inducible in response Inflammatory stimuli, including TNFα, IL-1, growth factors, bacterial endotoxin producing inflammation, pain sensitization, and fever response.
- REMEMBER: **Both COX-1 and COX-2** contribute to the generation of **autoregulatory** and **homeostatic** prostanoids, and both can contribute to prostanoid formation during **inflammation**.

How do the prostanoids exert their different effects in various body tissues?

- Prostanoids are autocrine and paracrine molecules that bind their specific GPCRs, activating intracellular signaling.
- Cells express specific prostanoid receptors according to their physiological functions.

Check your knowledge

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How do NSAIDs inhibit the synthesis of biologically active prostanoids?

- NSAIDs target COX-1 and COX-2, the enzymes involved in conversion of arachidonic acid to prostanoid precursors, thereby preventing the synthesis of prostanoids involved in many homeostatic and inflammatory functions.

How does the NSAIDs' mechanism of action relate to their therapeutic and adverse effects?

- The analgesic, anti-inflammatory, and antipyretic effects are due to inhibiting COX-2-derived prostanoid synthesis.
- Adverse effects occur due to disruption of the protective homeostatic functions of prostanoids, leading to potentially serious adverse effects in at-risk individuals.

Why is it helpful to recognize the affinities of the individual NSAIDs for COX-1 and COX-2?

- Less severe GI adverse effects occur the more COX-2 selective NSAIDs because the GI cytoprotective actions are mediated by COX-1-derived prostanoids.

Check your knowledge

How does NSAID use increase the risk of bleeding?

- Inhibiting COX-1-derived TXA2 synthesis in platelets reduces platelet aggregation and local vasoconstriction, which reduces formation of the platelet plug.
- How does use of nsNSAIDs and COX-2-selective drugs increase the risk of myocardial infarction and stroke?
- COX-2 selective agents inhibit endothelial COX-2-derived PGI2 synthesis, which reduces platelet disaggregation and local vasodilation, shifting the balance to TXA-2-mediated formation of thrombi, which can lead to myocardial infarction or stroke.
- nsNSAIDs inhibit synthesis of both platelet TXA2 and endothelial PGI2. COX-2 is more sensitive to NSAID inhibition, particularly under inflammatory/shear stress conditions → imbalance favoring TXA-2-mediated formation of thrombi, which can lead to myocardial infarction or stroke.

What organ system is most frequently injured by NSAID use?

- GI system: inhibition of COX-1-derived PGE2 and PGI2 inhibit mucus and bicarbonate secretion, blood flow, and inhibit the acid reducing effects of PGs.
  - Remember: NSAIDs reduce repair mechanisms and increase the risk of bleeding.

Check your knowledge

Which NSAIDs subclass has a dose-dependent lower risk of GI adverse effects?

- COX-2 selective agents do not inhibit COX-1-mediate cytoprotective effects.
- More COX-2 selective NSAIDs have lower affinity for COX-1-derived prostaglandins

Chronic or repeated NSAID use can lead to acute kidney injury by what mechanisms?

- Inhibiting the autoregulatory vasodilation of the afferent arteriole by PGE2 / PGI2 reduces renal perfusion and glomerular filtration rate.
- Patients with volume depletion, chronic kidney disease, heart failure, or cirrhosis are in states of sympatho-adrenal and renin-angiotensin-aldosterone upregulaton are at highest risk for AKI.

What are the various mechanisms of cardiovascular adverse effects?

- Inhibition of platelet TXA2 synthesis increases the risk of thrombotic myocardial infarction and stroke.
- Inhibition of renal Na+ and water excretion → increased Na+ and water reabsorption → increased blood volume, peripheral edema, and hypertension → increases blood pressure → the prothrombotic risk together with fluid retention can lead to myocardial infarction and stroke
- Hyperkalemia increases the risk of arrhythmia.

Check your knowledge

How does maternal chronic NSAID use affect the fetus?

- Oligohydramnios: Amniotic fluid is formed from fetal urine. From 20 weeks to term, reduced fetal renal function → reduced urine production → oligohydramnios → adverse fetal effects
- Closure of PDA: COX-2-derived PGE2 maintains the patent ductus arteriosus. NSAID use in the third trimester can potentially cause closure of the PDA → fetal pulmonary hypertension, right-sided heart failure, and fetal hydrops

Sum up the mechanisms of NSAIDs-related drug interactions in two sentences.

- Concurrent use of drugs that have the same effects (therapeutic or adverse effects) enhance NSAIDs-related adverse effects: GI irritants, drugs that increase bleeding (anticoagulants and antiplatelet agents), drugs that modify blood pressure and/or renal perfusion, drugs with CNS side effects
- Drugs that modify clearance of NSAIDs / NSAIDs that modify clearance of other drugs